

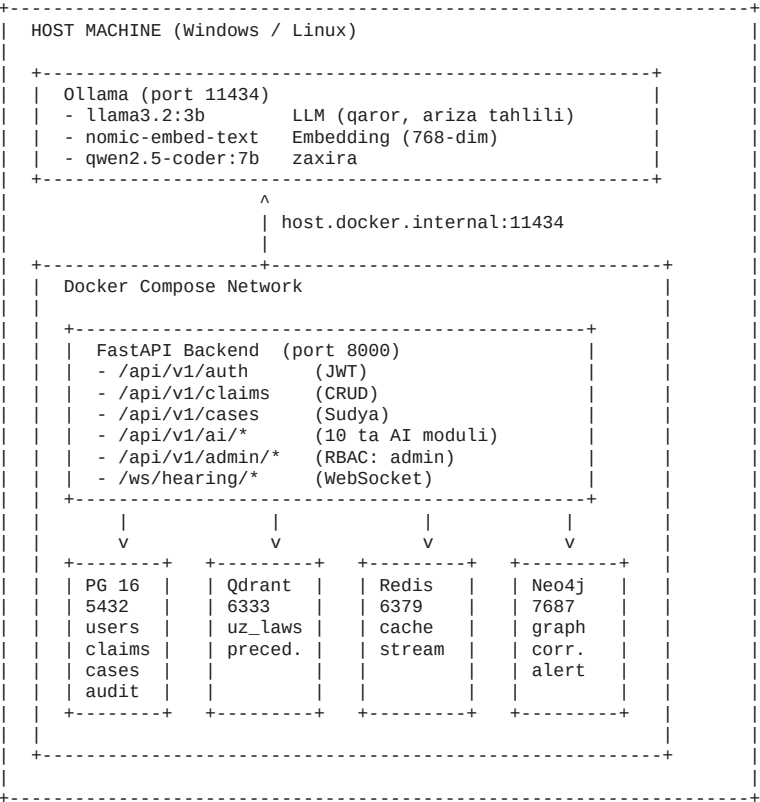
Texnik Arxitektura

SmartCourt AI — Backend & AI

Sana: 04.06.2026

1. Tizim sxemasi

SmartCourt AI mikroservis arxitekturasida qurilgan. Barcha AI modellari host'da (Ollama orqali) ishlaydi, qolgan barcha xizmatlar Docker konteynerlarda.

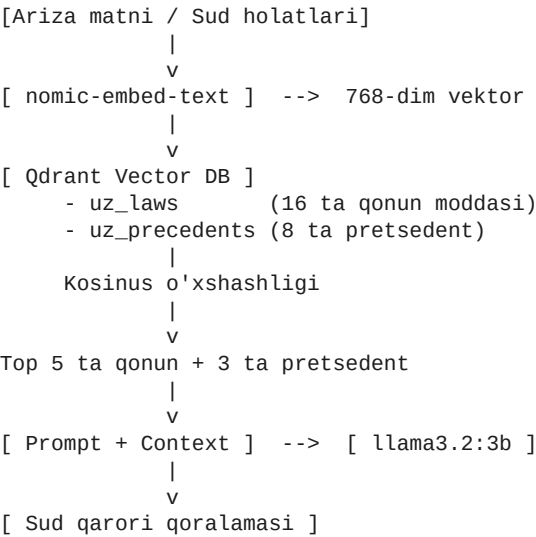


2. 10 ta AI moduli

Har bir modul alohida endpoint sifatida implementatsiya qilingan. RAG (Retrieval-Augmented Generation) qatlami SmartJudge, LexPredictor va Assistant tomonidan ishlatiladi.

#	Modul	Endpoint	Texnologiya
1	ClaimValidator	POST /ai/claim-validator	LLM + JSON output
2	MediatoBot	POST /ai/mediato-bot	LLM + JSON output
3	LexPredictor	POST /ai/lex-predictor	RAG (Qdrant + Embed)
4	EvidenceAnalyzer	POST /documents/upload	LLM + PDF/DOCX extract
5	JustiScribe	POST /justiscribe/transcribe + WS	faster-whisper
6	SentencAI	POST /ai/sentenc-ai	LLM + JSON scoring
7	SmartJudge	POST /ai/smart-judge/stream	RAG + LLM streaming
8	CorruptAlert	POST /ai/corrupt-alert/check	Neo4j graph
9	AnonimusLaw	POST /ai/anonimus-law	REGEX + NER
10	AutoExec	POST /ai/auto-exec/{id}	API integratsiya (mock)

RAG arxitekturası



3. Ma'lumotlar modeli

Jadval	Vazifa	Asosiy maydonlar
users	5 rol: citizen, judge, lawyer, admin, oversight	email, hashed_password, role, pinfl, court_name
claims	Fuqaro arizalari	id, claimant_id, dispute_type, title, description, amount, status, ai_validation, a
cases	Sud ishlari	id, reference, judge_id, decision_draft, decision_final, decision_signed_at, ai_
hearings	Sud majlislari (JustiScribe transcripti)	id, case_id, transcript JSONB, audio_path, ai_insights
documents	Dalillar, hujjatlar	id, claim_id, case_id, kind, path, mime_type, ai_analysis JSONB, extracted_te
notifications	Bildirishnomalar	id, user_id, title, body, is_read
audit_logs	Xavfsizlik audit	id, user_id, action, resource, ip_address, status, meta

Xavfsizlik qatlamlari

JWT	python-jose, HS256, 7 kunlik token
Parol	bcrypt (12 rounds), ochiq matn yo'q
RBAC	5 rol, require_roles() dependency
CORS	Whitelist orqali (CORS_ORIGINS)
Audit log	Har bir muhim harakat PostgreSQL'da yoziladi
PII Masking	AnonimusLaw — REGEX + NER, ochiq reestrga chiqishdan oldin
On-Premise	Ollama lokal — ma'lumot chet elga ketmaydi

4. API endpointlari ro'yxati

Jami 30+ ta endpoint. To'liq ro'yxat va interaktiv test — <http://localhost:8000/docs>

Auth & profile

```
POST /api/v1/auth/register
POST /api/v1/auth/login
POST /api/v1/auth/token (OAuth2)
GET /api/v1/auth/me
```

Citizen — Claims

```
POST /api/v1/claims
GET /api/v1/claims
GET /api/v1/claims/{id}
PATCH /api/v1/claims/{id}
POST /api/v1/claims/{id}/submit
DELETE /api/v1/claims/{id}
```

Judge — Cases

```
POST /api/v1/cases/from-claim
GET /api/v1/cases
GET /api/v1/cases/{id}
POST /api/v1/cases/{id}/sign
```

Documents (EvidenceAnalyzer)

```
POST /api/v1/documents/upload
GET /api/v1/documents/claim/{id}
```

AI modules (10)

```
POST /api/v1/ai/claim-validator
POST /api/v1/ai/mediato-bot
POST /api/v1/ai/lex-predictor
POST /api/v1/ai/evidence-analyzer/text
POST /api/v1/ai/sentenc-ai
POST /api/v1/ai/smart-judge
POST /api/v1/ai/smart-judge/stream (SSE)
POST /api/v1/ai/anonimus-law
POST /api/v1/ai/corrupt-alert/check
POST /api/v1/ai/auto-exec/{case_id}
POST /api/v1/ai/assistant
POST /api/v1/ai/assistant/stream (SSE)
POST /api/v1/justiscribe/transcribe
WS /ws/hearing/{case_id}
```

Admin (RBAC: admin)

```
GET /api/v1/admin/users
GET /api/v1/admin/stats
GET /api/v1/admin/audit-log
GET /api/v1/admin/ai-models
```

Dashboard

GET /api/v1/dashboard/citizen
GET /api/v1/dashboard/judge

System

GET /api/v1/system/health
GET /api/v1/system/status